MPJ Lab Assignment - 04

Sayali Patil
1032233901
26 (D1)

AIM :- Write a Java program to showcase the use of I/O File Exception Handling in Java.

PROBLEM STATEMENT :- Write a menu driven program for banking systems which accepts the personal data for Customers (cid, cname, amount). Implement the user-defined / standard exceptions, wherever required to handle the following situations.

1. Account should be created with a minimum amount of Rs. 1000.
2. For withdrawal of amount, if with_amt > total amount
3. cid should be in the specific range of 1 to 20
4. Entered amount should be positive.

Store these records in file using suitable I/O classes

OBJECTIVES :-

1. To study I/O stream classes
2. Understand the different types of exceptions
3. Exception Handling using various Exception classes.

THEORY :-

1] Java I/O stream classes

Java provides classes to perform input & output operations using streams

Types :-

- Byte Streams → Handle binary data
  Eg. - FileInputStream, FileOutputStream

- Character Stream → Handle text data
  Eg. - FileReader, FileWriter, BufferedReader

Exception Handling in Java

It is used to handle runtime errors & maintain normal program flow.

Keywords:

- try → Code that may cause exception
- catch → Handles exception
- finally → Always executes
- throw → Manually throw exception
- throws → Declare exception

Types of Exceptions:-

1. Checked Exceptions - checked at compile time
   Eg. - IOException
2. Unchecked Exceptions - occur at runtime
   Eg. - ArithmeticException
3. Userdefined Exceptions - Created by programmer
   Eg. - InvalidAmountException

FAQs:-

1] Difference b/w error & exception

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Error	Exception
• Serious problem in system	• Problem that occurs during program execution
• Cannot be handled by programmer.	• Can be handled using try-catch
• Unrecoverable	• Recoverable
• System level	• Application level
• OutofMemoryError	• ArithmeticException

2] Examples of Checked & Unchecked Exceptions

⇒ Checked :- • IOException
• SQLException
• FileNotFoundException

Unchecked :- • ArithmeticException
• NullPointerException
• ArrayIndexOutOfBoundsException

3] Different Keywords in Exception Handling

⇒ • try - contains code that may cause exception
• catch - handles exception
• finally - executes always (cleanup code)
• throw - used to explicity throw exception
• throws - declares exceptions in method signature

4] Difference b/w throw & throws.

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throw	throws
• used to throw an exception	• used to declare exception
• inside method body	• in method signature
• throw one exception at time	• can declare multiple exceptions
• throw new IOException();	• void fun() throws IOException

5] Byte Stream vs Character Stream Classes

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Byte stream	Character Stream
• Binary data	• Text data
• Byte (8-bit)	• Character (16-bit
• Images, audio file	• Text files
• FileInputStream, FileOutputStream	• FileReader, FileWriter